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Multi-Sensor Fusion for Autonomous Driving Perception

Abstract

Multi-sensor fusion has become a central design principle in autonomous driving perception because no single sensor simultaneously provides dense semantics, precise geometry, long-range robustness, velocity observability, and all-weather reliability. Cameras contribute rich appearance cues, LiDAR offers metrically accurate 3D structure, radar is resilient to adverse visibility and measures radial velocity, and emerging sensors such as event cameras and thermal cameras further extend the operating envelope. Over the last few years, the field has moved from relatively simple camera–LiDAR proposal fusion to unified bird's-eye-view fusion, query-based transformers, radar-centric fusion, and cooperative perception across connected agents. This survey synthesizes the modern literature on multi-sensor fusion for autonomous driving perception, emphasizing method families, representational choices, task scope, datasets, robustness, and open challenges. We argue that the historical evolution of the field can be understood through four recurring questions: where to fuse, what to align, how to represent uncertainty, and how to preserve robustness under sensing and communication failures. The survey also highlights a clear trend from task-specific pipelines toward unified perception backbones that support detection, segmentation, occupancy, map inference, and multi-agent scene understanding, while exposing unresolved problems in calibration, temporal consistency, domain generalization, safety, and evaluation. 1

Introduction

Perception for autonomous driving is inherently a multi-sensor problem. Urban road scenes are partially observed, dynamic, deformable, and frequently affected by range limitations, occlusion, lighting variation, weather, and sensor-specific failure modes. Cameras provide strong semantic discrimination but weak direct depth; LiDAR provides accurate 3D structure but sparse and texture-poor measurements; radar is inexpensive, long-range, Doppler-aware, and comparatively weather-resilient, but noisy and semantically weak. These complementary failure modes are precisely why fusion has become foundational rather than optional in high-performance driving stacks [1], [3], [4], [6], [7], [8]. 2

The development of fusion methods has been strongly shaped by benchmark availability. KITTI established the early camera–LiDAR regime for 3D detection; SemanticKITTI expanded dense semantic scene understanding over LiDAR sequences; nuScenes broadened the canonical sensor suite to six cameras, one LiDAR, and five radars with 360-degree coverage; Waymo Open Dataset increased scale and geographic diversity; ONCE emphasized large-scale 3D object detection; and newer datasets such as DAIR-V2X, OPV2V, TUMTraf-V2X, View-of-Delft, K-Radar, Seeing Through Fog, and DSERT-RoLL pushed the field toward cooperative perception, 4D radar, adverse weather, and richer multimodal combinations [1]-[12]. 3

The literature has now matured enough to support several survey-level syntheses. Recent reviews have categorized fusion by stage, representation, transformer design, cooperative topology, and radar-specific fusion mechanisms, while noting a consistent drift toward BEV-centric models, end-to-end differentiable alignment, and robustness-aware architectures [13]-[20]. Yet the literature remains fragmented: camera–LiDAR 3D detection dominates the mainstream, radar–camera fusion has accelerated rapidly, joint perception across vehicles and infrastructure is now a major subfield, and segmentation, panoptic perception, occupancy, and open-world fusion are expanding but still less consolidated. 4

This survey adopts a method-centered perspective. Instead of organizing the field only by task, we analyze it through representational and operational choices: raw-space fusion versus latent-space fusion, perspective-view versus point/voxel versus BEV fusion, object-centric versus scene-centric reasoning, single-agent versus cooperative fusion, and deterministic versus uncertainty-aware perception. This perspective is especially useful because the same design tensions recur across detection, segmentation, and cooperative perception. In each case, success depends on balancing geometric faithfulness, semantic richness, computational efficiency, synchronization tolerance, and robustness to sensor degradation [13]-[20], [39]-[69], [88]-[101]. 5

Methods

Sensor modalities, representations, and fusion taxonomies

The most durable taxonomy in the literature classifies fusion by representational stage. Early fusion injects one modality into another at the raw or low-level feature stage, for example by projecting image features onto points or augmenting voxels with camera semantics. Intermediate fusion combines learned modality-specific features at shared coordinates or latent tokens. Late fusion merges proposals, object hypotheses, or decisions. A second taxonomy classifies fusion by working space: image plane, range view, point set, voxel grid, BEV, or query/object space. A third taxonomy distinguishes scene-level fusion, which aims to form a coherent global latent representation, from instance-level fusion, which concentrates computation near object hypotheses [13], [15], [18], [19]. 6

These taxonomies are not merely descriptive; they directly encode trade-offs. Point- or voxel-level fusion maintains geometric fidelity but can be sparse, costly, and sensitive to calibration noise. Perspective-view fusion exploits dense image semantics but must bridge scale ambiguity and projection mismatch. BEV fusion compresses multi-sensor evidence into a geometry-friendly planning space, which is attractive for both perception and downstream control, but BEV construction itself may discard uncertainty and can be brittle when view transformation is inaccurate. Query-based fusion reduces computation by focusing on sparse object hypotheses, but risks losing background and contextual evidence that matter for small, occluded, or rare objects [39]-[56]. 7

A central insight from the modern literature is that fusion performance is limited less by the abstract choice of “modality combination” than by the quality of cross-modal alignment. Alignment is affected by calibration error, rolling-shutter or scan-time distortion, temporal offsets, motion parallax, differences in receptive field, and severe imbalance in sampling density. Many recent systems therefore place alignment modules at the core of the architecture rather than treating calibration as a preprocessing step. Examples include inverse augmentation and learnable cross-attention in DeepFusion, soft association in TransFusion, prior-guided alignment in Look Before You Fuse, and uncertainty-aware or failure-aware routing in more recent robust fusion models [39], [40], [51], [52], [69], [85], [86]. 8

LiDAR-camera fusion

The earliest generation of camera–LiDAR fusion largely operated through view aggregation or proposal refinement. MV3D [21] and AVOD [22] combined multiple geometric views for object proposal generation and refinement. Frustum PointNets [23] used 2D detections to crop frustum-aligned point subsets, demonstrating that image detections can sharply reduce 3D search space. PointFusion [24] and ContFuse [25] explored low-level cross-modal feature injection, while PointPainting [26], MVX-Net [27], 3D-CVF [28], EPNet [29], CLOCs [30], FusionPainting [31], and PointAugmenting [32] progressively improved methods for decorating points, voxels, or candidates with image semantics. This generation established a core lesson that still holds: even relatively simple semantic injection can improve long-range recall, object classification, and small-object detection, particularly when LiDAR is sparse [21]-[32]. 9

These methods were built on increasingly capable 3D backbones such as SECOND [33], PointPillars [34], PointRCNN [35], PV-RCNN [36], Voxel R-CNN [37], and CenterPoint [38]. The importance of these backbones in the fusion literature is not incidental. Fusion performance is governed not only by the fusion operator itself, but also by how well the underlying 3D encoder preserves localization cues, sparsity structure, and scale variation. Strong unimodal LiDAR baselines raised the bar substantially; as a result, multimodal methods needed more than simple concatenation to justify their complexity. This pressure was one of the forces that moved the field from feature painting toward object-aware attention and unified latent spaces [33]-[40]. 10

The next major phase was transformer- and BEV-centered fusion. TransFusion [39] used query-based LiDAR-camera fusion with soft association to improve robustness under degraded imagery. DeepFusion [40] addressed geometric augmentation mismatch through inverse augmentation and learnable cross-attention. FUTR3D [41] generalized the architecture into a sensor-agnostic query framework that could, in principle, absorb camera, LiDAR, and radar streams. UVTR [42], BEVFusion [43], UniTR [44], Virtual Sparse Convolution [45], SparseFusion [46], DeepInteraction [47], Cross Modal Transformer [48], and IS-Fusion [49] each explored different ways of performing fusion in shared latent space, ranging from voxel transformers to sparse candidate fusion and hierarchical instance–scene interaction. BEVFusion in particular marked an important conceptual step because it argued that the correct meeting place for multimodal perception is often a shared BEV rather than the point cloud itself, thereby preserving image semantics more effectively for detection and map-related tasks [39]-[49]. 11

Recent LiDAR-camera work has focused on three directions. First, supervision strategies such as SupFusion [50] attempt to explicitly regularize the fusion process so that the multimodal representation is not merely stronger, but also better aligned. Second, robust fusion models including Look Before You Fuse [51], MoME [52], and SAMFusion [53] introduce calibration-aware correction, sensor-conditioned expert routing, or adaptive sensor weighting to address sensor misalignment, weather degradation, and partial sensor failure. Third, self-supervised or weakly supervised approaches such as Self-Supervised Sparse Sensor Fusion [54] and related distillation-style frameworks are motivated by the cost of multimodal annotation at long range or in rare conditions. LoGoNet [55] and DeepInteraction++ [56] illustrate a broader theme in the field: fusion is increasingly treated as a problem of selective, object-aware information exchange rather than dense all-to-all blending [50]-[56]. 12

Radar-centric and adverse-weather fusion

Radar-centric fusion has grown quickly because radar is comparatively inexpensive and more robust to fog, rain, and snow than cameras or LiDAR. At the same time, radar is difficult to use well: point returns are sparse and noisy, angular resolution is limited, multi-path and ghost artifacts are common, and standard automotive radar outputs are often heavily preprocessed before reaching the learning system. These difficulties explain why radar–camera fusion lagged behind camera–LiDAR fusion for several years, even though radar is highly attractive for deployment [7], [8], [19]. 13

CenterFusion [57] was a pivotal early work because it demonstrated practical camera–radar fusion in a center-based detector and showed that radar can improve monocular 3D detection and tracking. More recent methods such as RCBEVDet [58], RaCFormer [59], ZFusion [60], R4Det [61], RPGFusion [62], IMKD [63], and RCTDistill [64] have shifted radar fusion toward BEV reasoning, query-based association, cross-view feature sampling, prior-guided depth conditioning, and teacher–student transfer. The modern radar–camera literature now treats radar not merely as an auxiliary cue but as a first-class structural prior that can help stabilize depth estimation, guide sampling, and improve performance under weak visual evidence [57]-[64]. 14

The emergence of 4D radar has further changed the design space. K-Radar [8] and View-of-Delft [7] showed that modern radar can be studied in benchmark settings with meaningful 3D supervision, while robust LiDAR–radar fusion methods such as Chae et al. [65], [66] exploit Doppler and weather resilience to compensate for LiDAR degradation. Temporal modeling is also especially natural for radar because Doppler and return persistence contain motion clues that are not well represented in static image features; Li et al. [67] demonstrated that temporal relational modeling can materially improve radar perception. Self-supervised radar learning, exemplified by Radical [68], is also important because radar data are abundant in fleet operation but expensive to annotate perfectly [65]-[68]. 15

A striking recent trend is the use of uncertainty-aware fusion for radar-rich settings. HyperDUM [69], for instance, explicitly models uncertainty contributions from multiple modalities and uses that information to guide fusion. This direction is significant because radar and camera failures are often condition-dependent, and robust deployment requires calibrated trust rather than fixed fusion weights. In practice, radar appears most valuable when models are trained not simply to “add radar,” but to reason about when radar should dominate, when it should be ignored, and how its uncertainty should interact with visual or LiDAR uncertainty [63], [64], [69]. 16

Fusion beyond single-frame single-vehicle detection

Although 3D object detection remains the dominant benchmark, multi-sensor fusion increasingly supports broader perception tasks. In semantic segmentation, xMUDA [70] used cross-modal learning for unsupervised domain adaptation; 2DPASS [71] demonstrated how strong 2D priors can assist LiDAR segmentation; MSeg3D [72] designed geometry-based feature fusion and cross-modal completion for multimodal 3D semantic segmentation; and LiDAR-camera panoptic segmentation methods such as Zhang et al. [73] moved toward joint semantic and instance reasoning. PanDA [74] extends this trajectory into multimodal 3D panoptic segmentation under domain shift. The major theme across these works is that segmentation needs both dense spatial support and reliable cross-modal correspondence; unlike object detection, it cannot hide alignment errors behind sparse object queries [70]-[74]. 17

Temporal and asynchronous sensing are also increasingly important. Ev-3DOD [75] fuses asynchronous events with synchronized sensors to push 3D detection to higher operational frequency, while DSERT-RoLL [76] provides a richer benchmark involving stereo event cameras, RGB, thermal cameras, dual LiDAR, and 4D radar. These works broaden the canonical sensor suite and suggest that future fusion systems will need to reason over heterogeneous time scales rather than assuming frame-synchronous inputs. In other words, the fusion problem is gradually becoming as much about temporal alignment and latency management as about spatial correspondence [75], [76]. 18

Another active direction is learning with limited labels and open-world supervision. VESPA [77] uses multimodal fusion together with vision-language models for open-vocabulary pseudo-label generation, while Reflective Teacher [78] studies semi-supervised multimodal 3D detection. Synthetic-to-real adaptation is also relevant: RadSimReal [79] addresses the gap between simulated and real radar, and Time to Shine [80] investigates synthetic adverse-weather fine-tuning. Together, these works indicate that the bottleneck in future fusion research may shift from architecture design toward label efficiency, open-set semantics, and cross-domain transfer [77]-[80]. 19

Cooperative and distributed fusion

Single-vehicle perception is fundamentally bounded by line-of-sight and onboard sensor placement, which has motivated cooperative perception. In cooperative settings, the fusion problem expands from sensor-level heterogeneity to agent-level heterogeneity, communication constraints, localization error, and adversarial risk. Early works such as Cooper [88], F-Cooper [89], and V2VNet [90] established the basic premise that collaborating agents can compensate for mutual occlusions and increase perception range. DiscoNet [91], V2X-ViT [92], and Where2comm [93] then developed stronger intermediate-fusion and communication-aware strategies, with Where2comm in particular showing that sparse communication policies can deliver favorable bandwidth–accuracy trade-offs [16], [88]-[93]. 20

The cooperative literature has now diversified in three directions. First, datasets such as DAIR-V2X [9], OPV2V [10], V2X-Seq [11], TUMTraf-V2X [12], and V2X-R [97] support increasingly realistic and multimodal evaluation. Second, architectural advances such as CORE [94], SparseAlign [95], and V2XPnP [96] emphasize sparse representation, spatio-temporal fusion, and efficient communication. Third, robustness-oriented models such as GSV2X [98] and the adaptive adversarial collaborative perception framework of Tao et al. [99] recognize that cooperative fusion is not automatically safe: it must cope with geometric inconsistency, malicious messages, and the fact that more information can also mean more failure modes [9]-[12], [94]-[99]. 21

Cooperative perception is also broadening beyond detection into occupancy and heterogeneous collaboration. Collaborative semantic occupancy prediction with hybrid feature fusion [100] shows that multi-agent fusion can support denser scene understanding than object lists alone, while the finding that collaboration can let camera-based systems rival or exceed LiDAR in certain settings [101] highlights an important systems insight: future fusion stacks may distribute sensing roles across infrastructure and fleets, rather than requiring every vehicle to carry the same premium sensor suite. This implies that multi-sensor fusion for autonomous driving must increasingly be studied jointly with deployment topology and communication policy, not only as a single-vehicle inference problem [100], [101]. 22

Challenges and Prospects

A first unresolved challenge is calibration and synchronization under realistic operating conditions. Most benchmarks assume reasonably accurate calibration, but deployment involves thermal drift, vibration, aging, maintenance variation, asynchronous clocks, and scan-time motion effects. The literature increasingly acknowledges that multimodal performance can degrade sharply under small alignment errors, especially at long range and object boundaries. Robust Long-Range Perception Against Sensor Misalignment [85] and Look Before You Fuse [51] are important because they treat misalignment as endemic rather than exceptional. A likely future direction is online calibration-aware fusion in which models estimate the trustworthiness of cross-modal correspondences jointly with the perception task itself [40], [51], [85]. 23

A second challenge is condition-dependent robustness. Adverse weather is not a single nuisance variable; it changes sensing physics differently across modalities. Fog attenuates image contrast and LiDAR return strength; rain and snow introduce clutter, scattering, and partial occlusion; night shifts the semantic burden toward active and thermal sensing. The most promising recent work therefore does not search for a universally dominant sensor, but for adaptive fusion policies. Seeing Through Fog [6], SAMFusion [53], Chae et al. [65], [66], MoME [52], and DSERT-RoLL [76] collectively point toward sensor-adaptive gating, weather-aware uncertainty modeling, and broader evaluation suites that include event and thermal cameras alongside radar and LiDAR [6], [52], [53], [65], [66], [76]. 24

A third challenge is uncertainty, safety, and security. Perception fusion is often informally justified as “more robust because more sensors are used,” but this is only partially true. Fusion can create overconfidence when correlated errors are incorrectly treated as independent, and it enlarges the attack surface for both physical and communication-based adversaries. HyperDUM [69] and Query2Uncertainty [86] show growing interest in calibrated confidence estimation, while Uncertainty-Guided Data Curation [87] suggests that uncertainty can also be used upstream for dataset management. At the same time, physical-world attacks on multimodal perception [82], latency attacks such as SlowLiDAR [83], and newer vulnerability studies [84], [99] show that redundancy does not automatically imply resilience. The field still lacks a widely adopted safety framework for sensor trust management, uncertainty propagation across modules, and failure-aware closed-loop evaluation [69], [82]-[87], [99]. 25

A fourth challenge is data efficiency and open-world generalization. High-quality multimodal labels are expensive, especially for long range, rare weather, infrastructure viewpoints, and radar-rich platforms. Semi-supervised learning, self-supervised radar learning, open-vocabulary auto-labeling, and simulation-to-real adaptation will therefore become increasingly central. The long-term value of datasets such as K-Radar, DAIR-V2X, V2X-R, and DSERT-RoLL may lie not only in benchmark score progression, but in enabling explicit study of how fusion behaves under sensor novelty, long-tail categories, and deployment shifts [8], [9], [76], [77], [79], [80]. 26

A fifth challenge is evaluation itself. Standard metrics such as mAP, NDS, and mIoU remain necessary, but they are insufficient for fusion. Fusion systems should also be compared under missing modalities, corrupted calibration, asynchronous sensing, constrained bandwidth, adversarial perturbation, and rare-condition subsets. Cooperative methods in particular need metrics that jointly account for perception accuracy, communication load, temporal consistency, and robustness to heterogeneous or malicious agents. The field is starting to move in this direction, but benchmark design still lags behind methodological ambition [16]-[20], [85]-[101]. 27

The prospects, nevertheless, are strong. The literature increasingly converges on several promising principles: unified latent representations such as BEV when suitable; sparse, object-aware fusion rather than dense indiscriminate blending; adaptive routing based on sensor trust; temporal fusion over asynchronous streams; self-supervised cross-modal pretraining; and cooperative perception that is communication-aware and security-aware from the outset. If these directions are pursued together, the future of autonomous driving perception is unlikely to be “sensor maximalism” in the sense of combining all available sensors all the time. More plausibly, it will be selective, uncertainty-aware, and deployment-aware fusion that uses the right information at the right time and degrades gracefully when sensing conditions worsen. [28]

Conclusion

Multi-sensor fusion for autonomous driving perception has progressed from heuristic cross-view aggregation to sophisticated, learned, and increasingly unified architectures spanning camera–LiDAR, camera–radar, LiDAR–radar, event-driven sensing, and cooperative multi-agent perception. The field’s central achievement has been to show that fusion is not simply an additive combination of sensors; it is a problem of geometric alignment, representational compatibility, trust calibration, and systems-level robustness. The strongest contemporary methods succeed because they fuse in spaces that preserve complementary information while explicitly addressing sparsity, uncertainty, and failure. At the same time, the field remains methodologically imbalanced: 3D detection is far more mature than segmentation, occupancy, or open-world fusion, and benchmark-driven progress still outpaces safety-oriented evaluation. The next phase of research is therefore likely to be defined by adaptive robustness, self-supervised and open-world learning, cooperative perception under communication and security constraints, and fusion policies that are aware of both sensing physics and downstream driving requirements. [29]

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